[**RL**] — The Reinforcement-Learning

1 Setup decisions that are physics, not hyperparameters

- **State representation** — inherit from [CH] §1.5.1 unchanged.
- **Reward** — $r_t \propto c(x_t)$ on the true field. Sparse global reward as a separate labelled experiment.
- **The discount factor γ is physics** — it sets a planning horizon $\tau_\gamma = \Delta t/(1-\gamma)$. If $\tau_\gamma \ll \langle T_{\rm esc}\rangle$ the agent is *provably indifferent* to the far peak, however much memory it has, so an apparent "failure to escape at deep $\Delta U_{\rm trap}$" may be an artifact of the objective rather than a fact about memory. Either hold $\tau_\gamma/\langle T_{\rm esc}\rangle$ fixed across the sweep, or promote γ to an explicit axis. *Recommended: make it an axis.*
- **Policy class** — add a softmax (Boltzmann) option alongside ε-greedy. The optimal finite-memory policy on a POMDP can be genuinely stochastic, and a greedy table cannot represent one — so "the 1-bit agent failed" could just mean "a deterministic table cannot express the answer."
- **Training-budget parity** — equal episodes, equal schedules across memory classes, or the hierarchy comparison measures compute rather than memory.
- **Exploration is not physics** — ε is training machinery; σ and D are the environment. All measurements on the frozen policy with exploration off.

2 The learnability experiment at $b = 1$

This is the smallest complete RL result, and it should be finished before anything else is attempted.

- **Train** — tabular Q-learning at one landscape setting and one σ, well inside the weak-signal window. Log the learning curve.
- **Freeze, then evaluate exactly** — take the frozen policy into [CH] and solve for $\langle T_{\rm esc}\rangle$ by one linear solve. No sampling error, no unescaped trajectories, no horizon truncation.
- **The learning gap** — compare against Module 2's exact optimum for the same memory budget. Report $\langle T_{\rm esc}\rangle^{\rm learned}/\langle T_{\rm esc}\rangle^{\rm optimal}$. *This single number is the result of Stage 1.*
- **Compare against the hand-coded policy** — the persist-or-reverse rule of [AD] §2.2 is a strong baseline. Does learning beat it, match it, or fall short?
- **Read the learned policy** — print the table. Does it reproduce persist-on-rise, reverse-on-fall? Where does it differ, and is the difference sensible?
- **Read the learned memory-update map f** — the most interpretable output the project can produce — what statistic did the agent decide to store? This is why the learner is tabular.

Gate

Do not proceed to Next section until the learning gap at $b = 1$ is a stable, reproducible number across seeds. If it is not, the problem is the learner (exploration, learning rate, γ, policy class), not the physics — and sweeping will only multiply the confusion.

3 Climbing the memory hierarchy

- **The independent optimum, per class** — exhaustive enumeration at $b = 0$ (a few hundred deterministic policies); CMA-ES or projected hill-climbing on the stochastic-policy simplex at $b = 1, 2$. Objective is [CH]'s exact MFPT. **This is not RL** and it is the more reliable route to the optimum.
- **Q-learning at $b = 0, 1, 2$ and window-m** — trained identically, benchmarked identically, gap reported per class.
- **The functional-form test, now for learned policies** — does the b=0 $\to$ b=1 transition from diffusive to Arrhenius survive when the policies are learned rather than hand-written? This is the project's headline claim, re-tested at the level that matters.
- **The committor diagnostic per class** — $\Delta q(i)$ from [CH] §5. Month 1 found $\Delta q \approx 0$ for the hand-coded 1-bit agent — memory converts comparison into drift but does not persist. **Does a learned b=2 agent behave differently?** If Δq becomes appreciable, memory has entered the reaction coordinate, and that is the first genuine non-Markovian signature.
- **Sensory vs. motor memory** — split the hierarchy along a second axis — retention of $\tilde{c}_{t-1}$ versus retention of a_{t-1}. Costs nothing to implement and sharpens the biological reading, since *E. coli* uses both.

4 Sweeps and the phase diagram

Only now, with a validated single-point pipeline, is a sweep worth its compute.

1. **Choose the axes and sweep in the right groups** — sweep the dimensionless combinations, not the raw parameters, or the data will not collapse and the "scaling law" will be parameterisation-dependent.
2. **The (σ, m) phase diagram** — regions: permanently trapped, intermittently escaping, reliably searching. Define the boundaries by a stated criterion, not by eye.
3. **The γ axis** — per §0.3. Reporting (γ, m) alongside (σ, m) is what makes the deep-barrier results interpretable.
4. **The escape-time surface** — $\langle T_{\mathrm{esc}} \rangle(\sigma, \Delta U_{\mathrm{trap}})$ per memory class. Remember: sweep ΔU_{trap} at *fixed* σ for slope extraction.
5. **Distributions, not means** — full first-passage histograms, with the fitting window taken from [CH]'s $\tau_{\mathrm{rel}} = -\Delta t / \ln |\lambda_2|$, stated explicitly.

5 The physics deliverables that genuinely need RL

1. **Effective-memory diagnostic** — $I(a_t; \tilde{c}_{t-k})$ per lag k. A window m is only an *upper bound* on memory used — the agent may ignore older entries. Plot scaling laws against *effective* memory, not nominal.

2. **The transmission factor** $\kappa(m)$ — measured by committor or reactive flux ([CH] §5), never fitted. Its dependence on memory is the Grote–Hynes payoff.

3. **Berg–Purcell comparison** — plot the RL-optimal memory window against the sensing bound SNR $\sim g v \tau_{\mathrm{mem}}^{3/2}/\sigma$. A match says the agent saturates the physical limit; a gap says something else is binding. Both are publishable.

4. **Rate-distortion frontier** $J^*(R)$ — achieved escape rate versus memory-information budget $R = I(M_t; \tilde{c}_{1:t})$. Monotone and concave by construction — verifying that is a check; locating the knee is the result.

5. **Landauer cost and the net-optimal budget** b^* — memory costs $\sim b k_B T \ln 2$ per update while the benefit saturates, so a finite b^* exists. Compare to the sub-bit-per-decision figure reported for real *E. coli*.

6 Reporting, and telling RL failure from physics

1. **The failure-mode checklist** — before calling any null result physical, rule out: τ_γ too short (§0.3); greedy policy class too weak (§0.4); unequal training budgets (§0.5); exploration left on during measurement (§0.6); Q-learning not converged on a POMDP.

2. **Report the learning gap everywhere** — every RL number is quoted alongside the exact optimum for the same class. This is what distinguishes a claim about *memory* from a claim about *Q-learning*.

3. **Negative results are results** — no sharp critical b_c, or a vanishing effect, is a genuine physical finding — plot it and defend it rather than burying it.

4. **Reproducibility** — JSON manifest per run: full parameter dict, seed, commit hash. At sweep scale, provenance is lost within a week otherwise.